

JKN RISK INTELLIGENCE

Detect Smarter. Investigate Faster. Protect JKN.

AI-powered healthcare risk intelligence that turns suspicious claims into **explainable, evidence-backed** investigations.



CATEGORY

Risk Efficiency in
Healthcare Facilities



RISK FOCUS

- Upcoding & Unbundling
- Phantom & Repeat Billing
- Inflated Bills & Cloning



Moch Aril Indra Permana
Team Lead

AI Engineer | Full Stack Developer
Machine Learning Enthusiast



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Member

Data Scientist | ML Engineer
Data Analyst

One Claim Can Hide Multiple Risks

A single claim may contain several suspicious indicators that require deeper investigation.

CLAIM SUMMARY



CLAIM #10293
RS Sehat Sentosa



Primary Diagnosis
Pneumonia berat



Severity Level
3



Length of Stay
6 days



Claim Amount
Rp18.450.000



Service Date
12 – 18 Mei 2026



Submission Date
19 Mei 2026

RISK SIGNALS DETECTED



1. UPCODING

Claimed severity level appears higher than supported by clinical evidence.

92%

HIGH



2. PHANTOM BILLING

Billed procedures lack sufficient supporting medical evidence.

81%

HIGH



3. ABNORMAL LOS

Length of stay is significantly longer than peer median for similar cases.

74%

MEDIUM



4. CLAIM SIMILARITY

High similarity with previous claims from the same provider.

68%

MEDIUM



“

The problem isn't just finding one anomaly. Investigators need to understand how multiple signals connect, what evidence supports them, and what to investigate next.

WHY THIS MATTERS



Multi-dimensional risk

One claim can trigger many suspicious indicators.



Context changes everything

Patterns, history, and relationships reveal the real story.



Human decision, AI support

AI helps investigators focus on the cases that truly need attention.



A suspicious claim is rarely just a score.
It is an investigation.

Why This Matters

Inefficient claims reduce the resources available for necessary care.

At the scale of JKN, even a small percentage of inefficient claims can create significant financial exposure and impact program sustainability.



275+
MILLION

Active JKN Participants

Serving more than 96% of
Indonesia's population



Rp165+
TRILLION

JKN Healthcare Spending (2023)

One of the largest health insurance
programs in the world



3-10%

Estimated Inefficiency Potential*
of Total Spending

Equivalent to Rp5-16 Trillion per year
that could be used for other services



Every rupiah lost due to inefficiency
is a service right lost for our participants.

Direct Impact of Inefficiency



Reduces funds for essential services
needed by participants.



Increases financial burden on the
national health system.



Hinders the sustainability of the JKN
program in the future.



Reduces trust in the integrity of
the system.

**Based on various international studies on health care fraud,
waste, and abuse.*

The Investigation Gap

Detecting an anomaly is only the beginning.

Investigators need to connect the dots, find the right evidence, and decide what truly matters.

CURRENT INVESTIGATION PROCESS



THE GAP



Too Many Signals

Investigators are overwhelmed by the volume of alerts.



Fragmented Evidence

Evidence is scattered across systems and formats.



Limited Context

Hard to see the full picture: patterns, relationships, and history.



Slow Prioritization

Right cases take too long to reach the right people.



We don't just need to detect more.
We need to investigate smarter.



Focus investigator capacity on the cases that matter most.

Meet JKN Risk Intelligence

From risk detection to evidence-backed investigation in one platform.

JKN Risk Intelligence is an AI-powered investigation platform that helps BPJS Kesehatan **detect**, **connect**, **explain**, and investigate suspicious claims more accurately and efficiently.



DETECT

Identify abnormal claims and emerging risk signals.



CONNECT

Reveal relationships across claims, providers, diagnoses, and procedures.



EXPLAIN

Turn risk signals into evidence-backed AI reasoning and insights.

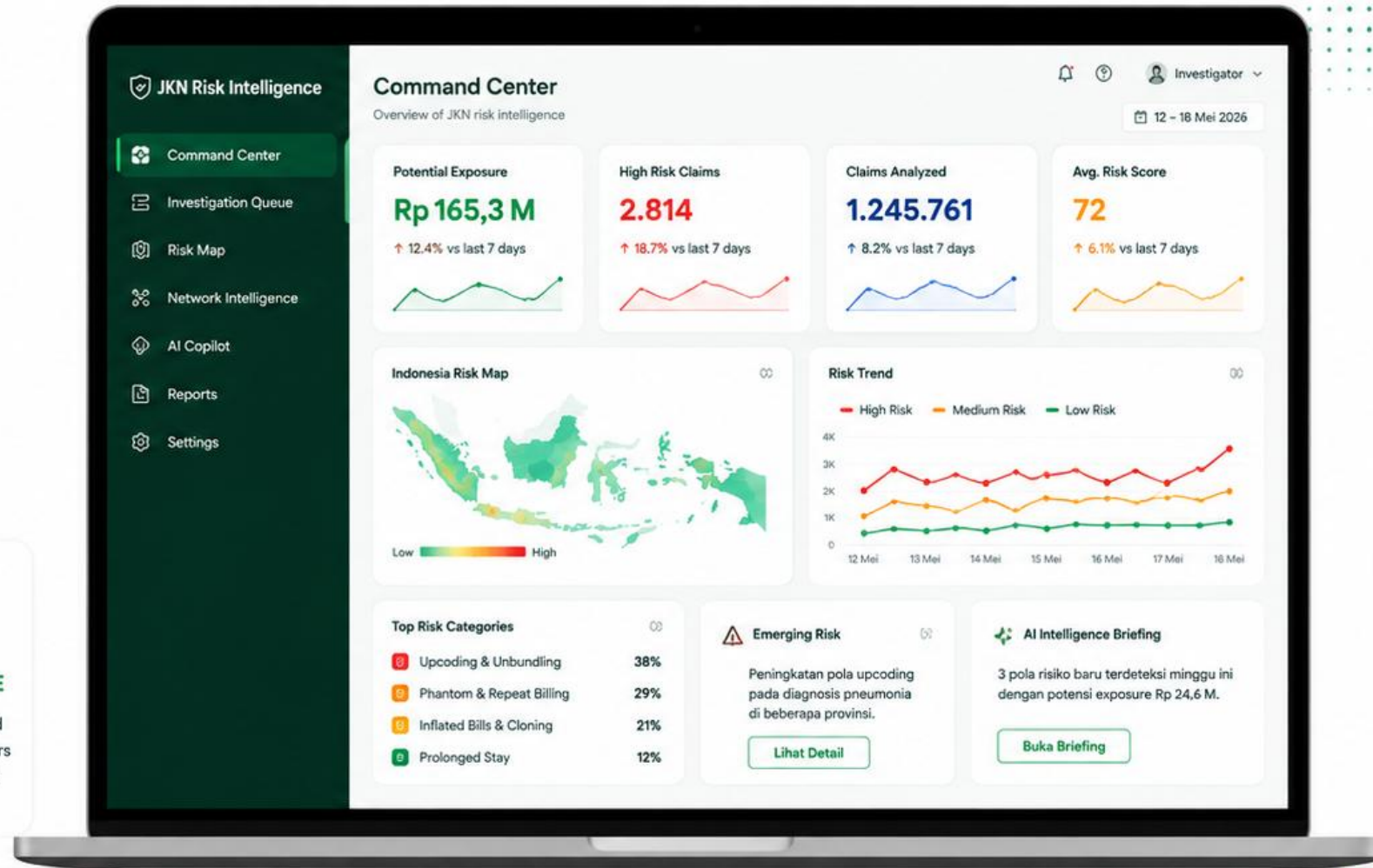


INVESTIGATE

Prioritize cases and support investigators to make confident decisions.



Not another dashboard.
An AI-assisted investigation platform.



One Platform, Multiple Risk Signals

Different risks. One investigation platform.

JKN Risk Intelligence combines multiple risk detectors in one unified platform so investigators can see the full picture and take the right actions.

01 UPCODING

Diagnosis / severity / billing consistency



Clinical Evidence

Claimed Severity / Billing



Detects

Inconsistencies between clinical evidence and claimed severity or billing level.

02 PHANTOM BILLING

Billed procedures without supporting evidence



Billed Procedure

Available Evidence



Detects

Procedures that are billed but not supported by medical records or documents.

03 CLAIM CLONING

Highly similar claims that may indicate duplicate or cloned billing



Claim A

Claim B



Detects

Unusually similar claims across patients, providers, or time periods.



EXTENSIBLE RISK ENGINE

Easy to add new detectors as new risk patterns emerge.

FUTURE DETECTOR MODULES



Repeat Billing



Abnormal Length of Stay



Unbundling



Inflated Bills



Readmission Pattern



Provider Anomaly



... and more



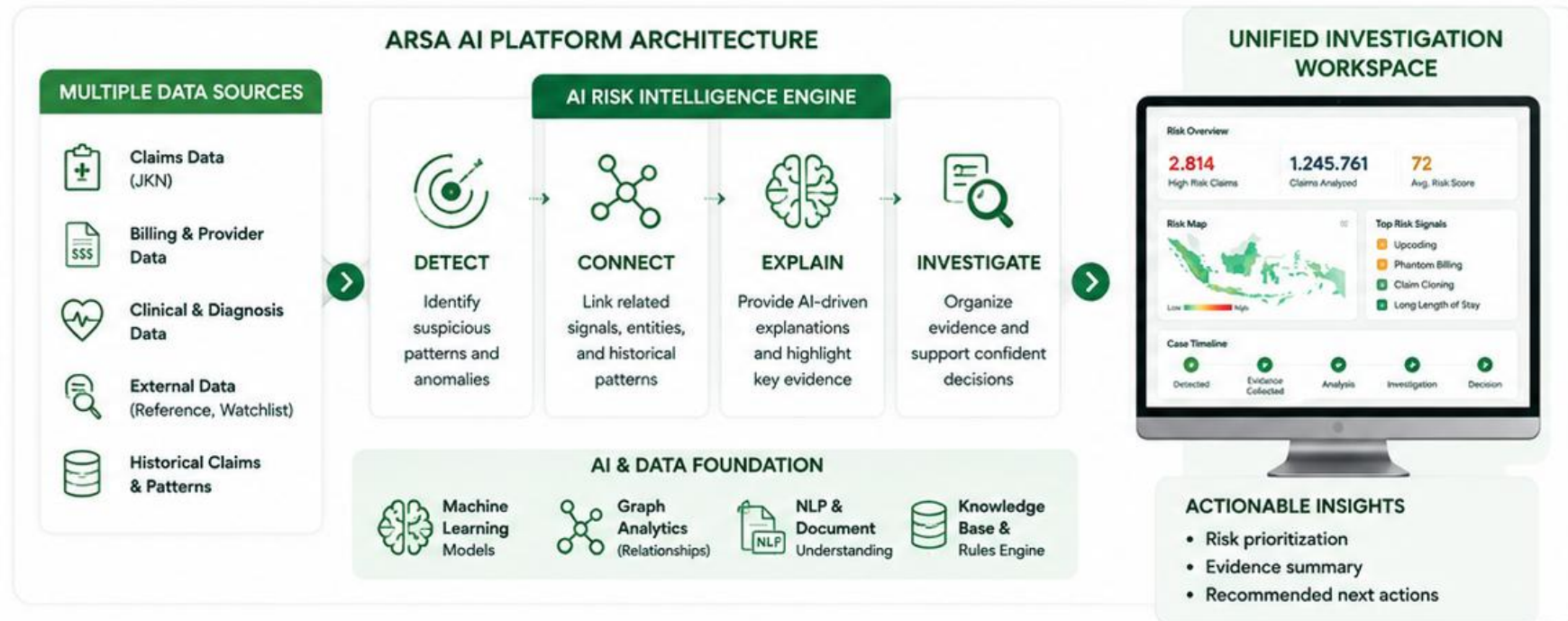
New risk types become detector modules,
not separate applications.

* All modules use synthetic data in prototype.

Our Solution: Arsa AI

AI-powered risk investigation platform that connects signals, evidence, and insights.

Arsa AI helps investigators detect, connect, explain, and investigate suspicious claims in one unified platform.



KEY BENEFITS



More Accurate Detection

Detect more risky claims with higher precision.



Connected Evidence

See the complete picture with connected data and context.



Faster Investigation

Reduce investigation time from days to minutes.



Better Decision

Make confident, consistent, and auditable decisions.

“

Arsa AI empowers investigators to focus on what matters most: the right cases, with the right evidence, for the right decisions.

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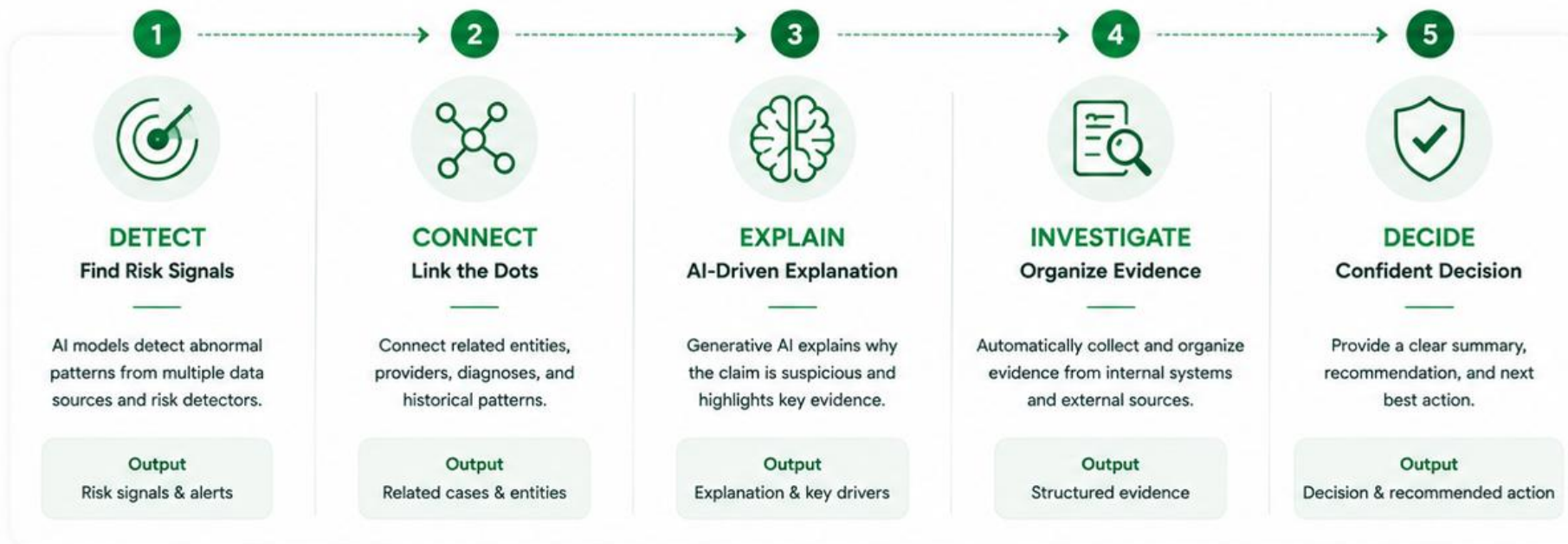


Arsa AI turns fragmented risk signals into clear insights, so investigators can act faster and smarter.

How Arsa AI Works

From risk signal to confident decision in 5 simple steps.

Arsa AI streamlines the investigation journey end-to-end, so investigators can focus on what truly matters.



BUILT FOR INVESTIGATORS

-  **Human-in-the-Loop**
Investigators stay in control with AI as a trusted assistant.
-  **Transparent & Explainable**
Every insight is backed by evidence and clear reasoning.
-  **Secure & Compliant**
Built with data privacy, security, and governance by design.
-  **Scalable & Adaptive**
Continuously learns from feedback and new risk patterns.



Arsa AI doesn't replace investigators.
It amplifies their capability to make better decisions.

Key Features of Arsa AI

Powerful capabilities to help investigators work smarter and faster.

Arsa AI brings together multiple risk detectors, connected data, and AI capabilities in an intuitive platform designed for real investigation workflows.



Unified Risk Radar

View all risk signals from multiple detectors in one prioritized dashboard.



Entity Connection Explorer

Visualize relationships between claims, providers, diagnoses, and entities.



AI Explanation Engine

Get AI-generated reasons, key drivers, and highlighted evidence for each case.



Evidence Organizer

Collect, organize, and annotate evidence from internal and external sources.



Investigation Workspace

End-to-end workspace to manage cases, tasks, notes, and collaboration.



Smart Case Prioritization

AI ranks cases based on risk level, potential impact, and available evidence.



AI Copilot for Investigators

Ask questions in natural language and get instant answers from your data.



Secure & Governed

Enterprise-grade security, role-based access, and full audit trail.



ANTICIPATED IMPACT



50-70%

Faster Investigation
Reduce investigation time from days to hours.



30-50%

More Accurate Detection
Identify more true risks with higher precision.



80%+

Better Evidence Completeness
Ensure all relevant evidence is collected and organized.



Higher

Investigator Productivity
Focus on high-value analysis and decision making.



Stronger

Governance & Compliance
Full traceability, auditability, and data security.



Arsa AI is more than a tool.

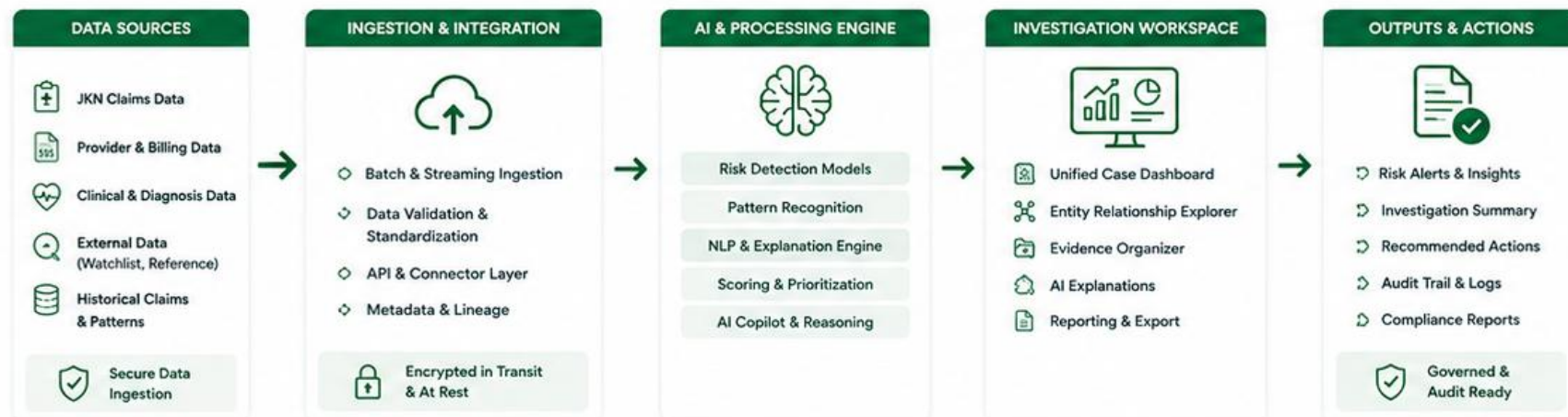
It's a force multiplier for investigators and a guardrail for the organization.

Technology & Architecture

Secure, scalable, and built for enterprise-grade investigations.

Arsa AI is built on modern cloud-native architecture with strong governance, security, and integration capabilities.

ARSA AI ARCHITECTURE OVERVIEW



PLATFORM FOUNDATION (CLOUD & SECURITY)



Google Cloud
(VPC, IAM, KMS)



BigQuery
(Data Warehouse)



Cloud Run
(Scalable Services)



Vertex AI / Gemini
(AI & LLM Services)



Monitoring & Logging
(Observability)



Zero Trust Security

RBAC & Least Privilege

Data Privacy & Compliance

Full Auditability



BUILT WITH GOVERNANCE IN MIND

- ✓ Zero Trust Architecture
- ✓ Role-Based Access Control (RBAC)
- ✓ Data Privacy & Compliance
- ✓ End-to-End Audit Trail
- ✓ Enterprise-Grade Security



SCALABLE & RELIABLE

- ✓ Cloud-native & serverless
- ✓ Auto-scaling for high volume
- ✓ High availability & disaster recovery
- ✓ Real-time monitoring & alerting



OPEN & INTEGRATED

- ✓ Flexible connectors & APIs
- ✓ Easy integration with existing systems
- ✓ Extensible for future detectors and capabilities



Arsa AI combines advanced AI with secure, scalable architecture to deliver trustworthy insights at enterprise scale.

Our Competitive Advantage

Arsa AI is purpose-built for healthcare risk investigation.

We combine AI, data, and domain expertise to deliver a platform that is more comprehensive, accurate, and actionable.



Unified Platform

All risk detectors and data sources in one integrated platform.



Complete visibility, faster discovery



AI-Powered Intelligence

Advanced AI/ML models that detect complex patterns and anomalies.



Higher accuracy, less false positives



Connected Evidence

Entity resolution and relationship mapping connect all the dots.



Stronger insights, better context



Built for Governance

Enterprise-grade security, audit trail, and compliance by design.



Trusted, compliant, and auditable



Investigator Centric

Intuitive workspace designed with real investigation workflows.



More productive, less manual work



Scalable & Future-Ready

Cloud-native architecture that grows with new detectors and data.



Adaptable today, ready for tomorrow



WHY IT MATTERS



Detect more true risks with higher precision.



Investigate faster and reduce backlog.



Ensure compliance and data security.



Empower investigators to focus on what matters.



Drive better outcomes and protect JKN sustainability.



Arsa AI isn't just another tool.

It's a strategic advantage for smarter, faster, and more impactful risk investigation.



50-70%

Faster Investigation

Reduce investigation time from days to hours.



30-50%

More Accurate Detection

Identify more true risks with higher precision.



80%+

Better Evidence Completeness

Ensure all relevant evidence is collected and organized.

Business Model & Sustainability

A sustainable model that creates value for BPJS Kesehatan and Indonesia.

Arsa AI is designed to deliver measurable impact today while building a long-term, sustainable ecosystem.

BUSINESS MODEL



SaaS Platform

Subscription-based access for BPJS Kesehatan investigators and authorized teams.

- Per user / role
- Tiered by usage & features
- Scalable pricing model



Integration Services

Professional services for integration with internal systems, data sources, and workflows.

- Discovery & assessment
- Implementation & onboarding
- Custom connector development



Advanced Analytics

Premium analytics, AI models, and risk intelligence for deeper insights and automation.

- Advanced risk scoring
- Predictive models
- Custom analytics reports



Training & Enablement

Trainium programs, and knowledge transfer to build internal capability and adoption.

- User training & certification
- Best practice workshops
- Continuous enablement



REVENUE GROWTH ROADMAP



MVP & Pilot

Validate value, refine solution, and prove impact.

2026



Early Adoption

Expand to more teams and use cases within BPJS Kesehatan.

2027



Scale & Expansion

Enterprise-wide adoption and advanced analytics expansion.

2028



Ecosystem Growth

Open partnerships and ecosystem collaboration.

2029+



A future-ready investment for measurable impact, sustainable growth, and national health protection.

VALUE & IMPACT



Operational Impact

Reduce investigation time by 50-70% and improve investigator productivity.



Quality Impact

Improve detection accuracy by 30-50% and reduce false positives.



Financial Impact

Prevent potential fraud and improper claims, protecting JKN fund sustainability.



Strategic Impact

Strengthen data-driven decision making and organizational capability.



Sustainability Impact

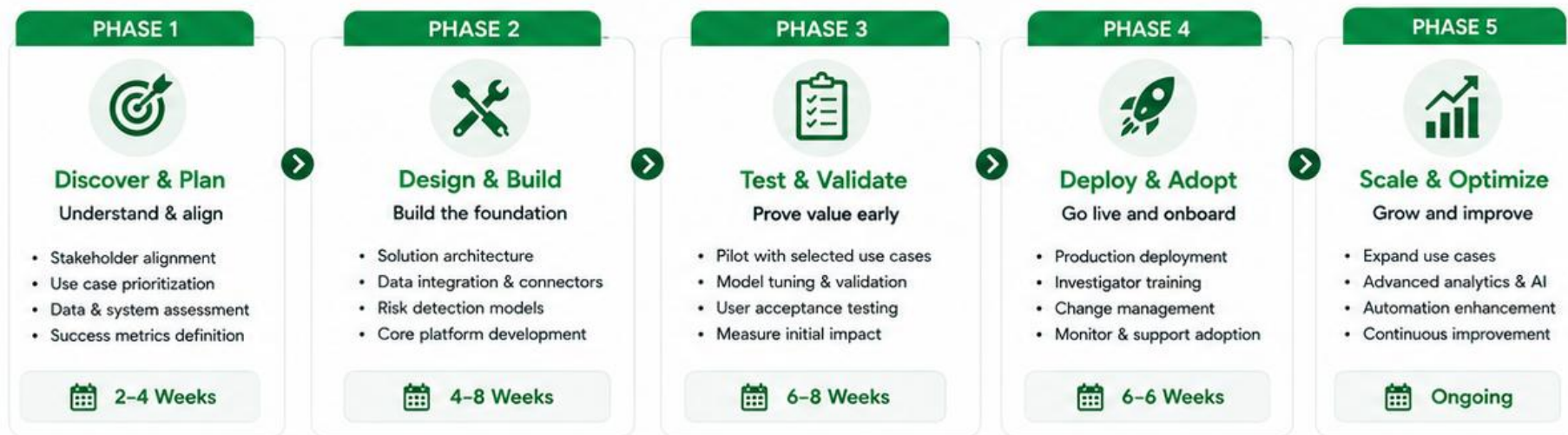
Build a scalable, secure, and future-ready platform for long-term value.

“ Arsa AI creates sustainable value by combining technology, people, and process to **protect the JKN program** and **empower a healthier Indonesia**. ”

Implementation Plan

A practical roadmap to build, validate, and scale Arsa AI.

We follow an agile and phased approach to ensure quick wins, continuous learning, and sustainable adoption.



KEY MILESTONES



KEY SUCCESS FACTORS

-  **Strong Collaboration**
Close partnership across business, IT, data, and governance teams.
-  **Data & Security First**
Privacy, security, and compliance built in from day one.
-  **User-Centered Adoption**
Designed for investigators, with training and ongoing support.
-  **Measure & Iterate**
Track impact, learn fast, and continuously improve.
-  **Start Small, Scale Fast**
Deliver quick wins, then scale with confidence.

“

A clear plan. Measurable milestones. Sustainable impact.
Arsa AI is built to deliver results—today and for the future.

Budget & Resource Plan

Efficient investment for maximum impact.

We allocate resources strategically to build a high-impact, scalable, and sustainable solution.

BUDGET BREAKDOWN (ESTIMATED)

CATEGORY	DESCRIPTION	ESTIMATED COST (IDR)	PERCENTAGE
 Cloud Infrastructure	GCP services (Cloud Run, BigQuery, Storage, VPC, IAM, Monitoring)	70,000,000	28%
 AI & Data Services	Vertex AI, Model training & tuning, AI APIs, Data processing	80,000,000	32%
 Development & Integration	Platform development, connectors, integrations, and testing	60,000,000	24%
 Security & Compliance	Security tools, audit, penetration test, compliance & governance	25,000,000	10%
 Training & Enablement	Investigator training, documentation, workshops, change management	15,000,000	6%
TOTAL ESTIMATED BUDGET		250,000,000	100%

RESOURCE PLAN



People

Cross-functional team with AI, data, engineering, and domain experts.



Technology

Leverage GCP, ADK, MCP, and modern open standards.



Tools

CI/CD, monitoring, observability, collaboration, and security tools.



Partnerships

Collaboration with BPJS Kesehatan, domain experts, and technology partners.

INVESTMENT SUMMARY



Total Investment

250,000,000 IDR

Strategic investment for high impact and sustainability.



Expected ROI

High Impact

Faster investigations, accurate detection, and cost efficiency.



Time to Value

6 – 8 Weeks

First measurable results from pilot phase.



Long-term Value

Sustainable Growth

Scalable platform for continuous improvement and innovation.



Smart investment today for a stronger, more efficient, and sustainable investigation ecosystem tomorrow.

Team & Call to Action

The right team, the right mission, the right impact.

We are passionate about leveraging AI and data to create meaningful impact for Indonesia's healthcare system.

OUR TEAM



Moch Aril Indra Permana
Ketua

AI Platform Engineer with expertise in MLOps, cloud architecture, and AI solution development.

- ✓ 8+ years in data & AI
- ✓ GCP, MLOps, & Data Engineering
- ✓ Platform & Solution Architect



Salfarisa Fitria
Anggota

Data Scientist with strong analytical skills and experience in AI modeling and data analysis.

- ✓ Data Science & Analytics
- ✓ NLP, ML, & Statistical Modeling
- ✓ Insight Driven & Detail Oriented

PROJECT TIMELINE



Problem Discovery & Research

Aug 2026

Deep dive into user pain points and data exploration.



Solution Design & Prototype

Sep 2026

Design architecture, build prototype, and validate approach.



Development & Integration

Oct 2026

Build core features, integrate data sources, and ensure security.



Testing & Validation

Nov 2026

Test accuracy, reliability, and user acceptance.



Final Demo & Presentation

Dec 2026

Deliver final solution and demonstrate impact.

OUR TARGET IMPACT (12+ MONTHS)



50-70%

Faster Investigations
Reduce investigation time from days to hours.



30-50%

More Accurate Detection
Identify more true risks with higher precision.



80%+

Better Evidence Completeness
Ensure all relevant evidence is collected and organized.



100+

Investigators Empowered
Equip more investigators with AI-powered insights.



Let's build a smarter, faster, and more trustworthy healthcare ecosystem together.

Vote for Arsa AI. Impact for Indonesia.



CALL TO ACTION

We believe Arsa AI can transform healthcare risk investigation in Indonesia.
We are ready to build it. We are ready to make an impact.

Why Arsa AI, Why Now?

The right solution, at the right time, for a better Indonesia.

Healthcare risk investigation is complex and time-consuming.

Arsa AI brings a new way forward—intelligent, connected, and impactful.

TODAY'S CHALLENGES



Manual & Time-Consuming
Investigations take days or weeks, delaying decisions.



Inconsistent Accuracy
Human bias and limited data lead to missed risks.



Fragmented Data
Data silos make it hard to get a complete picture.



Resource Constraints
Investigators are overloaded with repetitive tasks.



Fraud & Leakage Risk
Potential fraud and improper claims impact JKN sustainability.



ARSA AI SOLUTION



AI-Powered Automation
Reduce investigation time by 50-70% from days to hours.



Higher Accuracy
Advanced AI models detect more true risks with higher precision.



Unified & Connected Data
Integrate multiple data sources for 360° risk visibility.



Empowered Investigators
Focus on critical cases, not repetitive manual tasks.



Stronger Governance
Built-in security, compliance, and audit to protect JKN.

THE IMPACT WE CREATE



50-70%
Faster Investigations
Deliver results in hours, not days.



30-50%
More Accurate Detection
Identify more true risks with higher precision.



80%+
Better Evidence Completeness
Ensure all relevant evidence is collected and organized.



100+
Investigators Empowered
Enable more investigators to focus on what matters.



Sustainable Impact
Protect JKN sustainability and build a healthier Indonesia.

Thank You for Your Time and Consideration

Arsa AI is more than a solution.

It is a commitment to build a smarter, faster, and more sustainable healthcare ecosystem for Indonesia.



Together, we can make healthcare smarter, investigations faster, and Indonesia healthier.
Let's build the future, together.



Team Arsa AI
Smarter Investigation. Healthier Indonesia.



arsa-ai.id



hello@arsa-ai.id



Arsa AI



Jakarta, Indonesia

SLIDE 19

Thank You!

Detect Smarter. Investigate Faster.

We appreciate your time, attention, and support.
Together, let's build ethical AI that protects people
and creates a healthier Indonesia.



Protect

Safeguard people
and sensitive data



Empower

Empower investigators
with trustworthy AI



Impact

Drive real impact
with data & AI



Sustain

Build a sustainable
future together



Arsa AI is our commitment to **protect people**,
preserve trust, and **create lasting impact**.

